

Software Requirements Specification (SRS)

Project: LegalEase AI – Indian Legal Document Assistant

1. Purpose

LegalEase AI is a web application that helps users understand legal documents such as employment contracts, rental agreements, NDAs, loan agreements, and service contracts by using an AI agent to analyze uploaded PDFs and generate simple explanations, risk assessments, and summaries.

2. Objectives

- Upload legal PDFs
- Extract and classify documents
- Detect clauses and risks
- Explain clauses in plain English
- Generate summaries and downloadable reports
- Maintain user history

3. Stakeholders

End Users, Hackathon Judges, Developers, Administrators.

4. Functional Requirements

FR1 User authentication.

FR2 PDF upload.

FR3 Text extraction (PyMuPDF).

FR4 AI document classification.

FR5 Clause segmentation.

FR6 Risk scoring.

FR7 Plain-language explanation.

FR8 Overall summary.

FR9 Download PDF report.

FR10 View previous analyses.

FR11 Responsive dashboard.

5. Non-Functional Requirements

Response under 20 seconds for typical documents, secure storage, HTTPS, responsive UI, scalable API, accessibility.

6. System Architecture

Frontend: Next.js + Tailwind CSS

Backend: FastAPI

Database: Supabase PostgreSQL

AI: OpenAI API

Deployment: Vercel + Render

7. AI Agent Workflow

Upload PDF → Extract Text → Classify Document → Split Clauses → Analyze Clauses → Risk Score → Explain → Summarize → Save History → Generate Report

8. Database

Users(id,name,email), Documents(id,user_id,file,type,text),

Analyses(id,document_id,risk,summary), Clauses(id,analysis_id,title,risk,explanation).

9. API Endpoints

POST /upload

POST /analyze

GET /history

GET /analysis/{id}

GET /download/{id}

10. UI Modules

Landing Page, Login, Dashboard, Upload, Analysis View, History, Settings.

11. Security

JWT authentication, encrypted transport, file validation, role-based access, prompt sanitization.

12. Testing

Unit tests, API tests, UI tests, AI output validation, performance testing.

13. Future Enhancements

Hindi support, contract comparison, chat with document, OCR for scanned PDFs, clause highlighting, email reports.

14. Hackathon Deliverables

Deployed app, public GitHub repository, 3-minute demo, Google Doc, README, architecture diagram.